Human Intention Modeling for Physical Human–Robot Co-Manipulation

Control-relevant representations, inference methods, and an initial experimental study

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Research Discussion with RISE Lab

Human intention is hidden but control-relevant

What the robot observes

- object pose and velocity x_t^O;
- interaction wrench $\mathbf{w}_t^{\mathrm{int}}$ (3D force + 3D torque);
- human/robot motion and recent actions;
- task geometry, goal, and constraints c.

What remains hidden

- desired object motion or local goal;
- leader/follower role and compliance;
- effort, force, and safety preferences;
- responses to possible robot actions.

Central question

How should human intention be represented and inferred so that it improves real-time robot planning and control in physically coupled tasks?

A useful model must resolve four uncertainties

- **Behavioral multimodality**: the same state may admit several valid goals, paths, or roles.
- **Partial observability**: motion and force provide indirect, sometimes conflicting evidence.
- **Closed-loop interaction**: the human responds to the robot and may change strategy.
- **Bounded rationality**: actions may be delayed, noisy, effort-limited, or non-optimal.

Terminology

"Multimodal sensing" means multiple input types (motion, wrench, vision). "Multimodal behavior" means multiple plausible future human strategies. The two are different.

Shared papers show human models benefit interaction

Diffusion Co-Policy

- Generates multiple plausible joint force/action sequences
- Demonstrates adaptation and role switching
- Provides the physical test setting for collaborative transport

Bounded-Rational Model

- Shows joint human actions can be represented by a game model
- Bounded rationality fits observed behavior better
- Leadership and adaptation affect performance

PACE / N-PACE

- Estimates intent/cost parameters online
- Treats the peer as another adapting learner
- Shows a reasonable peer model can improve interaction policies

Combined insight

Human intention is modelable and useful for control; the unresolved question is which representation is sufficiently expressive, identifiable, and transferable.

Five control-relevant intention representations

Interaction primitive

Output: discrete mode and continuous parameters $z_H = (m, g, v, \rho)$.

Goal / desired object trajectory

Output: a terminal object pose, path mode, or desired trajectory ξ_H^*.

Cost and preferences

Output: features ϕ and weights w_H describing progress, effort, force, and safety tradeoffs.

Local dynamical system

Output: a human-intended vector field $f_H(x^O; \theta_H)$ over object motion.

Interactive response distribution

Output: possible interaction-wrench/action responses conditioned on candidate robot actions.

Selection criteria

Identifiability, predictive accuracy, real-time control utility, and transfer across tasks.

High-level option: Learn reusable interaction primitives

Switching latent-state model

$$z_t^H = (m_t, g_t, v_t, \rho_t), \qquad m_t \in \mathcal{M}_{\text{int}}$$

where $\mathcal{M}_{\text{int}}$ includes translate, rotate, hold, align, insert, and yield.

- m_t: current interaction primitive;
- g_t: local direction, pose, or contact goal;
- v_t: desired speed or interaction-force level;
- ρ_t: leader/follower role or compliance.

Why it supports generalization

Different tasks reuse a small vocabulary: transport may be *translate–rotate–align*; insertion may be *approach–align–hold–insert*.

How it can be trained

Use weak task-phase labels when available and semi-supervised EM / state-space learning otherwise. Train continuous parameters from future motion and wrench prediction.

Related idea: ADACORL learns factored human decision models with limited labels (Unhelkar et al., CoRL 2020).

Option A: Predict the human's desired object motion

Desired trajectory model

$$p_\psi(\xi_H^* \mid h_t, c), \qquad \xi_H^* = \{x_{O,k}^{H,*}\}_{k=t}^{t+H}$$

h_t contains recent object/human/robot motion and interaction wrench; c contains task goal, geometry, object, and constraints.

Possible outputs

- terminal pose or waypoint;
- discrete path/strategy mode;
- distribution over full desired trajectories.

Why it is attractive

Directly useful as a reference for shared-object MPC; easier to label and visualize than latent cost parameters.

Main challenges

Multiple valid trajectories; changing intention; passive prediction ignores robot influence.

Training without a unique intent label

Use future object motion as a self-supervised target, but learn a distribution rather than one "correct" path. Optional goal labels come from controlled conditions; diffusion or best-of-K losses preserve multiple valid futures.

Best viewed as a **low-level control output**, not a complete human model (Townsend et al., 2017; Franceschi et al., 2023).

Option B: Infer what the human is optimizing

Task-conditioned latent cost

$$J_H = \sum_{k=t}^{t+H} w_k^{H^\top} \phi_\psi(x_k^O, u_k^H, u_k^R, c)$$

- ϕ_ψ: shared learned and physical features for progress, effort, force, safety, and coordination;
- w_H: person-specific preference weights inferred online.

Why it is attractive

Interpretable tradeoffs and a direct connection to game-theoretic planning, IOC, PACE, and N-PACE.

Main challenges

Non-identifiability, semantic drift in learned features, and approximate human optimality.

Training without cost labels

Learn shared ϕ_ψ from demonstrated actions via max-entropy IOC/IRL with physical-feature anchoring; infer person-specific w_H as a latent online variable. No direct cost labels are required.

The trained cost model can become N-PACE's human model

Offline learned representation

- Train a shared task-conditioned feature map ϕ_ψ from multi-person demonstrations.
- Learn a context-dependent prior $p(w_H \mid c)$.
- Freeze ϕ_ψ at deployment so the online parameter meaning remains stable.

Online N-PACE layer

- Define $J_H = \sum_k w_H^\top \phi_\psi(x_k, u_k^H, u_k^R, c)$.
- N-PACE estimates w_H and how the human is updating beliefs about the robot.
- The inferred cost enters the dynamic-game solver to update robot actions.

What the learned model contributes

It replaces task-by-task manual feature design with a reusable cost representation; N-PACE retains peer-aware online inference and interaction-aware control.

Option C: Model intention as a local vector field

Human-intended dynamical system

$$\dot{x}_O^{H,*} = f_H(x_O, c; \theta_H)$$

For a simple attractor model:

$$\dot{p}_O^{H,*} = K_p(p_H^* - p_O), \qquad \omega_O^{H,*} = f_R(R_O, R_H^*; K_R)$$

$\theta_H = (p_H^*, R_H^*, K_p, K_R)$ represents desired pose and response aggressiveness.

Why it is attractive

No assumption that the human solves a complete optimal-control problem; parameters can be updated from local motion and interaction-wrench evidence.

Difference from Option B

Option B explains behavior as approximate horizon optimization of a cost. Option C directly models local desired velocity; it is lighter-weight but less expressive about long-term tradeoffs.

Training without parameter labels

Fit object-motion likelihood by constrained system identification. Infer θ_H and switching modes with EM or a particle filter; impose stability and feasibility on f_H.

Related: constraint-aware DS intent estimation with particle filtering (Shao et al., 2024).

Option D predicts responses to candidate robot actions

Passive prediction

$$p(u^H_{t:t+H} \mid h_t, c)$$

What will the human do next?

$\Longrightarrow$

Interactive response model

$$p_\psi(u^H_{t:t+H} \mid h_t, c, u^R_{t:t+H})$$

How will the human respond if the robot executes this candidate action?

- u^H: future interaction-wrench/action sequence; u^R: candidate robot action sequence.
- A diffusion/sequence model can represent multiple responses and role switches.
- It is predictive, but intention remains implicit unless the model also exposes a goal, trajectory, or preference variable.

Option D versus Diffusion Co-Policy

Diffusion Co-Policy

$$p_\psi(u^H_{t:t+H}, u^R_{t:t+H} \mid h_t, c)$$

- jointly generates human and robot action sequences;
- directly imitates collaborative demonstrations;
- captures multimodality and role switching.

Human response model

$$p_\psi(u^H_{t:t+H} \mid h_t, c, \tilde{u}^R_{t:t+H}, z_H)$$

- evaluates each candidate robot action $\tilde{u}^R$;
- keeps planning and safety modular;
- can expose an intent variable z_H.

Tradeoff

The response model supports counterfactual planning, but needs diverse robot-action coverage and calibrated out-of-distribution uncertainty.

Each representation serves a different modeling layer

Representation	Control interface	Key strength	Main weakness	Best use
Interaction primitive	Latent mode z_H	Cross-task structure	Needs mode discovery	Task-phase inference
Desired trajectory	Reference ξ_H^*	Direct and visual	Misses preferences and robot influence	Shared-object MPC
Cost / preferences	J_H, w_H, ϕ	Interpretable and game-compatible	Identifiability and feature design	Negotiation and tradeoffs
Dynamical system	$f_H(x; \theta_H)$	Lightweight online inference	Limited global multimodality	Local coordination
Response distribution	$p(u_H \mid u_R)$	Flexible and interactive	Intention may remain implicit	Candidate-action evaluation

Proposed position

Use structured intention variables as the control interface, while a learned response model captures multimodality, bounded rationality, and model mismatch.

A simpler hierarchy connects intention modeling to control

Observations motion + interaction wrench + context → Interaction primitive $z_H = (m, g, v, \rho)$ → Control-relevant intention $\xi_H^* \; / \; f_H(\theta_H) \; / \; w_H^\top \phi_\psi$ → Human response model $p(u_H \mid h, c, \tilde{u}_R, z_H)$ → N-PACE / MPC robot action u_R

Learned offline

Primitive transitions, shared feature/trajectory representations, and the population response prior.

Adapted online

Current primitive, desired motion, preference weights, and person-specific response uncertainty.

Offline priors support online personalization

Offline: learn shared structure

- task/interaction encoders from multi-person data;
- trajectory modes and residual cost features;
- robot-action-conditioned response model;
- calibrated uncertainty on held-out people.

Online: update compact variables

- belief over intent mode $b_t(z_H)$;
- desired motion parameters ξ_H^*, θ_H;
- preference weights w_H;
- optional small person-specific adapter.

Update from physical interaction

$$b_{t+1}(z_H, \theta_H, w_H) \propto p(\mathbf{w}_t^{\mathrm{int,obs}}, x_{t+1}^O \mid h_t, c, u_t^R, z_H, \theta_H, w_H)\, b_t(z_H, \theta_H, w_H)$$

Data collection supplies weak and self-supervision

Minimum synchronized modalities

- object pose, velocity, and acceleration;
- 6D interaction wrench;
- robot state and commanded action;
- task goal, geometry, contact configuration;
- optional human body/arm pose and vision.

Where training signals come from

- **weak labels**: assigned goal, path condition, task phase, role instruction;
- **self-supervision**: future object motion and interaction wrench;
- **preference evidence**: choices under controlled effort/force/path tradeoffs;
- **latent learning**: EM / IOC likelihood when no direct label exists.

Practical collection plan

Use accessible human–human demonstrations for pretraining, then collect human–robot trials with diverse robot actions for interactive-response identification. A curated multimodal benchmark can itself support a dataset/benchmark paper.

First study: Compare intention representations

Physical test setting

Collaborative object transport with:

- two feasible paths around an obstacle;
- translation–rotation tradeoffs;
- changing leader/follower roles;
- synchronized object state and 6D interaction wrench.

Model comparison

1. desired-trajectory prediction;
2. fixed-feature IOC / cost inference;
3. switching dynamical-system inference;
4. interactive response model;
5. hierarchical hybrid model.

Central hypothesis

Combining structured intention variables with robot-action-conditioned response prediction will improve early intent inference and closed-loop coordination.

Separate model quality from team performance

Human-model quality

- path/role classification and detection delay;
- desired-trajectory displacement error;
- force/action likelihood and calibration;
- adaptation after strategy changes;
- leave-one-person-out generalization.

Closed-loop interaction quality

- task success and completion time;
- human and robot effort;
- unnecessary interaction force;
- path efficiency and smoothness;
- safety and comfort violations.

Critical ablations

Motion vs. motion+wrench; passive vs. robot-action-conditioned; fixed vs. learned features; population prior vs. online personalization.

A focused decision can define the first paper

Proposed contribution

Compare control-relevant human intention representations for physical collaborative transport, then combine their complementary strengths in a hierarchical model.

Questions for discussion

- Which output is most useful for the intended controller: desired object motion, cost/preferences, or future interaction wrench?
- Which intentions can be manipulated or labeled reliably in the planned testbed?
- Should the initial paper emphasize model comparison or one hybrid implementation?
- What robot, object, and force/torque sensing setup is already available?

Recommended starting scope

Begin with desired-trajectory and cost/preference baselines, then add robot-action-conditioned response prediction as the learned component.

Follow-up Research Directions

Longer-term subprojects in sensing, control, and cross-task generalization

SP1: Build the physical evidence base

Theme and focus

Physically grounded multimodal data

- Build shared-object transport experiments
- Synchronize motion, object state, and force/torque
- Create ambiguous paths, role switching, and strategy changes
- Compare motion-only, wrench-only, and multimodal inference

Challenges → solutions

- No direct intent labels → hierarchical task/strategy/action labels
- Noisy contact data → filtering and contact-state estimation
- High person variability → probabilistic latent-state models
- Costly data → human–human, simulation, and HRI data mixture

Expected paper: A multimodal benchmark and systematic study of force/torque for human intent inference.

SP2: Make intent and preference explicit

Bounded-rational human model

$$J_H = \sum_t w_t^{H\top} \phi_\psi(x_t, u_t^H, u_t^R, c)$$

Reusable features may represent progress, pose, effort, interaction force, safety, smoothness, and coordination.

- Infer intent, role, preference, and rationality online
- Represent the human's belief about robot intent
- Treat the human as another adapting learner

Challenges → solutions

- Cost identifiability → Bayesian / maximum-entropy IOC
- Ambiguous intent → informative robot actions
- Unknown adaptation → latent learning-state estimation
- Online computation → local game approximations and MPC

Expected paper: Peer-aware intent and preference inference for physical co-manipulation.

SP3: Learn responses to possible robot actions

Interactive response prediction

$$p_\psi(u^H_{\text{future}} \mid h_t, c, u^R_{\text{future}})$$

- Predict multimodal human response trajectories
- Condition on motion, wrench, context, and candidate robot actions
- Predict uncertainty and latent-state transitions
- Compare Transformer, diffusion, and latent-variable models

Challenges $\rightarrow$ solutions

- Limited action coverage $\rightarrow$ diverse robot probing
- Correlation vs. response $\rightarrow$ counterfactual conditioning
- OOD actions $\rightarrow$ uncertainty estimation and detection
- Slow inference $\rightarrow$ distilled or few-step diffusion

Expected paper: Robot-action-conditioned diffusion models for multimodal human response prediction.

SP4: Close the loop through safe control

Human-model-aware planning

$$u_R^* = \arg\min_{u_R} \mathbb{E}_{u_H \sim p_\psi}[J_R]$$

subject to object dynamics, actuation, force, collision, and safety constraints.

- Optimize progress, team effort, and interaction force
- Replan as the human model is updated
- Use a safety filter before execution

Challenges → solutions

- Prediction error → short-horizon MPC
- Distributional complexity → scenario sampling
- Excess conservatism → risk-sensitive objectives
- Hard safety limits → CBF / constrained safety filter

Expected paper: Uncertainty-aware planning and safe control with interactive human response models.

SP5: Transfer human concepts across tasks and people

Task-conditioned generalization

Shared human latent concepts:

- intent and local strategy;
- leader–follower role and compliance;
- effort, safety, and interaction preferences;
- reaction delay and adaptation state.

Encode goals, object dynamics, geometry, contacts, and constraints as task context. Adapt only latent states, preference weights, or a small adapter online.

Challenges → solutions

- Different task structures → object-centric states and wrenches
- Limited physical data → simulation and human–human pretraining
- Negative transfer → structured task encoders
- Unstable personalization → Bayesian updates or small adapters

Expected paper: A task-conditioned human model for cross-task co-manipulation and online personalization.

Five subprojects answer one connected question chain

SP1 — Observe: What physical signals reveal human intent and interaction strategy?

↓

SP2 — Infer: What does the human intend, prefer, and believe about the robot?

↓

SP3 — Predict: How will the human respond to candidate robot actions?

↓

SP4 — Act: How should the robot coordinate safely under prediction uncertainty?

↓

SP5 — Transfer: What representations generalize across people and tasks?

A focused transport study can test the core hypothesis

Initial task

A human and robot transport a shared object through an environment with multiple feasible paths.

Model inputs

- Object and human motion
- Interaction force/torque
- Goal and environment geometry
- Candidate robot actions

Comparisons

1. Reactive control without a human model
2. Passive prediction from interaction history
3. Structured intent inference
4. Robot-action-conditioned prediction
5. Hybrid structured + learned model

Evaluation

- Prediction accuracy and calibration
- Success, time, and team effort
- Interaction force and disagreement
- Adaptation, safety, and comfort violations

The program connects human understanding to robot action

1. A physically grounded multimodal benchmark
2. Bounded-rational, peer-aware intent inference
3. Interactive prediction of responses to robot actions
4. Safe planning over human response distributions
5. Cross-task generalization and online personalization

Long-term vision

A versatile human model that connects physical interaction, human adaptation, and robot decision-making across co-manipulation scenarios.

Proposed starting point

Use a physical collaborative-transport task to test whether force/torque sensing and robot-action-conditioned prediction improve intent inference, coordination, and safe control.

Questions to refine the research direction

- Which physical co-manipulation task is the most useful initial test case?
- What sensing and robotic platforms are currently available?
- Should the first contribution emphasize human intent inference, interactive response prediction, or planning and control?
- Which components of RISE Lab's existing game-theoretic framework can be extended directly to physical coupling?
- What level of cross-task generalization would be most valuable for the current project?

Discussion objective

Identify a focused first study that contributes to the larger goal of a generalizable and interactive human model.